Complexity Cheat Sheet

Data Structures and Algorithms - Time and space at a glance

Data structure operations (average case)

Structure	Access	Search	Insert	Delete
Array	O(1)	O(n)	O(n)	O(n)
Linked list	O(n)	O(n)	O(1)*	O(1)*
Stack / Queue	-	O(n)	O(1)	O(1)
Hash table	-	O(1)	O(1)	O(1)
BST (balanced)	-	O(log n)	O(log n)	O(log n)
Binary heap	-	O(n)	O(log n)	O(log n)

** Insert/delete at a known position (head, or given node in a doubly linked list).*

Sorting algorithms

Algorithm	Best	Average	Worst	Stable
Bubble	O(n)	O(n^2)	O(n^2)	yes
Insertion	O(n)	O(n^2)	O(n^2)	yes
Selection	O(n^2)	O(n^2)	O(n^2)	no
Merge	O(n log n)	O(n log n)	O(n log n)	yes
Quick	O(n log n)	O(n log n)	O(n^2)	no
Heap	O(n log n)	O(n log n)	O(n log n)	no

Graph algorithms

- BFS / DFS: O(V + E).
- Dijkstra (binary heap): O((V + E) log V).
- Bellman-Ford: O(V * E), handles negative edges.
- Kruskal: O(E log E). Prim (heap): O(E log V).